

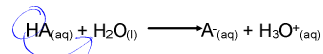
Arrhenius Definitions

There are different definitions of what makes acids and bases.

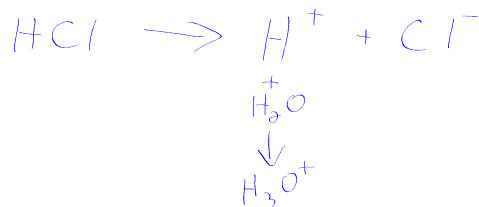
First, we will be looking at the Arrhenius definitions.

Arrhenius Acid:

Produces a hydrogen ion (H^+) or hydronium (H_3O^+) when dissolved in water. The general formula is:

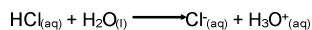


HA is a polar covalent molecule that contains a proton (H^+) that easily dissociates and bonds with a water molecule.

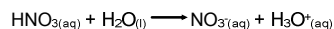


Examples

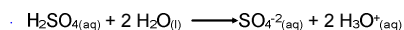
Hydrochloric Acid - HCl



Nitric Acid - HNO_3



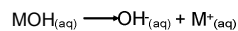
Sulfuric Acid - H_2SO_4



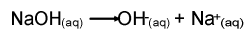
Arrhenius Base:

Produces a hydroxide ion (OH^-) when dissolved in water. There are two different ways this can happen:

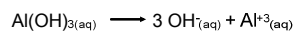
- 1) A hydroxide ion dissociates from an ionic compound:



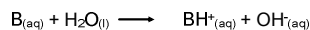
Ex) Sodium hydroxide - NaOH



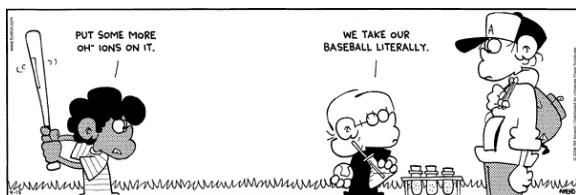
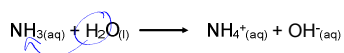
Ex) Aluminum hydroxide - $Al(OH)_3$



- 2) A covalent molecule steals a proton from a water molecule:



Ex) Ammonia - NH_3



Read p454 - 457
Do p457 #1, 2, 4, 6